

IRC—Internet Relay Chat

Jarkko Oikarinen started Internet Relay Chat (IRC) in 1988 and revolutionised the chat system. It was an implementation that allowed people who were logged onto a Bulletin Board Service (BBS) to communicate with each other. People with special interests could come together online for the first time.

The use of IRC saw a massive growth when Kuwait was invaded by Iraq in 1991. Users from all over the world wanted to have a first-hand glimpse of what

was happening in Kuwait. When all other lines of communication were cut, an IRC link to Kuwait was framed, which remained active for more than a week. For a glimpse of what happened, check out www.ibiblio.org/pub/academic/communications/logs/Gulf-War/.

Since then, instant messaging has moved down an evolutionary path to what is available today from various service providers.

message is handed over to the chat client. The client identifies the address of the user from the directory on the communication server and routes it to him/her. (See table, 'Instant Messengers Compared')

That's the underlying principle of how you can send a message from India to a friend in the US within seconds. But what makes instant messaging better than e-mail? Usually an e-mail has to travel through numerous servers before it reaches the recipient and hence is slower. In the case of instant messengers, however, a message is routed through only one server (addresses of your buddies are stored on the network of communication servers across the world) and the message reaches the recipient who is connected to the same server, instantaneously. The time lag between the message relay and arrival is so infinitesimal that you find instant messaging almost instantaneous. Besides the speed factor, e-mail also lacks the interactive and personal touch that an IM offers.

Video, voice and file transfers

Your IM is a collaborative tool that not only lets you type in text niceties but also lets you see and hear your loved ones. The

with an authentication server. The server searches its vast database for the username and password combination that you keyed in (see diagram, 'A working chat session'). If the search is successful the server lets you know that you have been successfully connected.

Instant Messaging: Once you are authorised to use the chat service via a particular client, a connection is established with another server called the communication server. The client sends your computer's address (IP + port) to the communication

server. The server maintains a directory of users who are logged onto it, which includes the user ID and IP. In return the server sends your buddy list to the IM client. The client frequently sends packets of information to the server inquiring about the status of your buddies and when one of them signs in, goes offline or changes his status, your client is informed with an appropriate signal from the server.

Once you key in a message to your friend and hit the Send button, your

INFOGRAPHICS: Jaya Shetty

A Working Chat Session

Authentication server: contains the data for all user IDs and passwords. It gives you the authorisation ticket to log on to the communication server.

Communication server: All user IDs and IPs (mobile numbers if any) are mapped in the directory. Once a user sends a message, the communication server looks up the IP and relays the message to the same.

- 1 a. Server verifies username and password
b. Server sends an authorisation token
- 2 a. Send IM to be relayed to a friend
b. Receive message through the communication server
c. Reply to sender
d. Receive response through communication server
- 3 a. File, voice or video connection request
b. Accept or reject token

